

```

> restart: unprotect('D'):

m := 6;

twist := - 6;

d := 16;          degree := d + 1:

##

HUU := X^(d+1) + T^(d+1) + 1*X^floor((d+1)/2)*T^(d+1-floor((d+1)
/ 2));

##

UU := subs({T=1,X=x}, HUU);

##

Jetyx := expand(

+ (1/Rz^(m-0*2)) * add(AA||j * x1^(m-0*3-j) * (y1)^j *
Delta_yx^0, j=0..m-0*3)

+ (1/Rz^(m-1*2)) * add(BB||j * x1^(m-1*3-j) * (y1)^j *
Delta_yx^1, j=0..m-1*3)

+ (1/Rz^(m-2*2)) * add(CC||j * x1^(m-2*3-j) * (y1)^j *
Delta_yx^2, j=0..m-2*3)

+ (1/Rz^(m-3*2)) * add(DD||j * x1^(m-3*3-j) * (y1)^j *
Delta_yx^3, j=0..m-3*3)

) :

##

dA := (m-0*2)*d - m + twist:

dB := (m-1*2)*d - m + twist:

dC := (m-2*2)*d - m + twist:

dD := (m-3*2)*d - m + twist:

```

##

Delta_yx := y1*x2 - y2*x1:

Soly1 := - x1*Rx/Ry - z1*Rz/Ry:

Soly2 := -x1^2*Rxx/Ry+2*x1^2*Rxy*Rx/Ry^2+2*x1*Rxy*z1*Rz/Ry^2-2*x1*z1*Rxz/Ry-Ryy*x1^2*Rx^2/Ry^3-2*Ryy*x1*Rx*z1*Rz/Ry^3-Ryy*z1^2*Rz^2/Ry^3+2*z1*Ryz*x1*Rx/Ry^2+2*z1^2*Ryz*Rz/Ry^2-z1^2*Rzz/Ry-x2*Rx/Ry-z2*Rz/Ry:

##

Jetzx := sort(mtaylor(

expand(subs(x2=(Deltazx+x1*z2)/z1,
expand(subs({y1=Soly1,y2=Soly2},

Jetyx))))),

[x1,z1,Deltazx], 100), [x1,z1,Deltazx]):

##

printlevel := 2:

for k from 0 to m/3 do

for j from 0 to m-3*k do

EEEq[m-j-3*k,j,k] := factor(

Rz^(m-j-3*k)

*

Ry^(m-2*k)

*

coeftayl(Jetzx, [x1,z1,Deltazx]=[0,0,0], [m-j-3*k,j,k])):

od: od:

EQUATION RR EXPLICITE

RR := z^(d+1) + y^(d+1) + UU;

Rx := diff(RR,x):

Ry := diff(RR,y):

Rz := diff(RR,z):

Rxx := diff(RR,x,x):

Rxy := diff(RR,x,y):

Rxz := diff(RR,x,z):

```

Ryy := diff(RR,y,y):
Ryz := diff(RR,y,z):
Rzz := diff(RR,z,z):

```

```
##
```

```

printlevel := 2:
for k from 0 to m/3 do
for j from 0 to m-3*k do

```

```
EEEq[m-j-3*k,j,k] := mtaylor(EEEq[m-j-3*k,j,k], z, (m-j-3*k)*d):
```

```
od: od:
```

```
##
```

```
for j1 from 0 to 1+m*d+d do
```

```
ny[j1] := (-z^(d+1)-UU)^(floor(j1/(d+1))) * y^(j1-(d+1)*floor(j1/
(d+1))):
```

```
od:
```

$$\begin{aligned}
m &:= 6 \\
\text{twist} &:= -6 \\
d &:= 16 \\
HUU &:= T^{17} + T^9 X^8 + X^{17} \\
UU &:= x^{17} + x^8 + 1 \\
RR &:= x^{17} + y^{17} + z^{17} + x^8 + 1
\end{aligned}$$

(1)

```
> ##
```

```
for l from 0 to m-0*3 do AA||l := add(A||l[j]*y^j, j=0..min(d,
dA)): od:
```

```
for l from 0 to m-1*3 do BB||l := add(B||l[j]*y^j, j=0..min(d,
dB)): od:
```

```
for l from 0 to m-2*3 do CC||l := add(C||l[j]*y^j, j=0..min(d,
dC)): od:
```

```
for l from 0 to m-3*3 do DD||l := add(D||l[j]*y^j, j=0..min(d,
```

```

dD)): od:

##

1 + m*d + d:

printlevel := 2:
for k from 0 to m/3 do
for j from 0 to m-3*k do

degree(EEEq[m-j-3*k,j,k], y):

EEEq[m-j-3*k,j,k] := mtaylor(EEEq[m-j-3*k,j,k], y, 1+m*d+d):

dy[m-j-3*k,j,k] := degree(EEEq[m-j-3*k,j,k], y):

od: od:

##

printlevel := 3:
for k from 0 to m/3 do
for j from 0 to m-3*k do

for j1 from 0 to dy[m-j-3*k,j,k] do

Cy[m-j-3*k,j,k,j1] := coeff(EEEq[m-j-3*k,j,k], y, j1):

od:

od: od:

##

printlevel := 2:
for k from 0 to m/3 do
for j from 0 to m-3*k do

EEq[m-j-3*k,j,k] := add(Cy[m-j-3*k,j,k,j1]*ny[j1], j1=0..dy[m-
j-3*k,j,k]):

degree(EEq[m-j-3*k,j,k], y):

od: od:

##

printlevel := 2:

```

```

for l from 0 to m-0*3    do
for j from 0 to min(d,dA) do

```

```

A||l[j] := add(A||l[j,k]*z^k, k=0..dA-j):

```

```

od: od:

```

```

printlevel := 2:
for l from 0 to m-1*3    do
for j from 0 to min(d,dB) do

```

```

B||l[j] := add(B||l[j,k]*z^k, k=0..dB-j):

```

```

od: od:

```

```

printlevel := 2:
for l from 0 to m-2*3    do
for j from 0 to min(d,dC) do

```

```

C||l[j] := add(C||l[j,k]*z^k, k=0..dC-j):

```

```

od: od:

```

```

printlevel := 2:
for l from 0 to m-3*3    do
for j from 0 to min(d,dD) do

```

```

D||l[j] := add(D||l[j,k]*z^k, k=0..dD-j):

```

```

od: od:

```

```

##

```

```

printlevel := 2:
for k from 0 to m/3    do
for j from 0 to m-3*k do

```

```

EEq[m-j-3*k,j,k] := mtaylor(EEq[m-j-3*k,j,k], z, (m-j-3*k)*d):

```

```

degree(EEq[m-j-3*k,j,k], z):

```

```

od: od:

```

```

##

```

```

printlevel := 3:
for l from 0 to m-0*3 do
for j from 0 to min(d,dA) do
for k from 0 to dA-j do

A||l[j,k] := add(A||l[i,j,k]*x^i, i=0..dA-j-k):

od: od: od:

```

```

printlevel := 3:
for l from 0 to m-1*3 do
for j from 0 to min(d,dB) do
for k from 0 to dB-j do

B||l[j,k] := add(B||l[i,j,k]*x^i, i=0..dB-j-k):

od: od: od:

```

```

printlevel := 3:
for l from 0 to m-2*3 do
for j from 0 to min(d,dC) do
for k from 0 to dC-j do

C||l[j,k] := add(C||l[i,j,k]*x^i, i=0..dC-j-k):

od: od: od:

```

```

printlevel := 3:
for l from 0 to m-3*3 do
for j from 0 to min(d,dD) do
for k from 0 to dD-j do

D||l[j,k] := add(D||l[i,j,k]*x^i, i=0..dD-j-k):

od: od: od:

```

##

```

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;

```

##

```

printlevel := 2:
for k from 0 to m/3 do
for j from 0 to m-3*k do

Eq[m-j-3*k,j,k] := mtaylor(EEq[m-j-3*k,j,k], z, (m-j-3*k)*d):

degree(Eq[m-j-3*k,j,k], z):

od: od:

##

printlevel := 2:
for k from 0 to m/3 do
for j from 0 to m-3*k do

degree(Eq[m-j-3*k,j,k], y):

od: od;

```

51255

51255

51255

51255

51255

51255

51255

17799

17799

17799

17799

1751

16

16

16

16

16

16

- ∞

16

16

16

- ∞

- ∞

(2)

```
> ## SYSTEM FOR DIVISIBILITY BY  $z^{(1*d)}$ 

Systemez1d := {}:

for k from 0 to (m-1)/3 do

  Coeffs_z1d[1,m-1-3*k,k] := {coeffs(expand(mtaylor(Eq[1,m-1-3*k,
k], z, 1*d)), [x,y,z])}:

  Systemez1d := Systemez1d union Coeffs_z1d[1,m-1-3*k,k]:

od:

Systeme_z1d := Systemez1d:

## VARIABLES FOR DIVISIBILITY BY  $z^{(1*d)}$ 

Variables_z1d := indets(Systeme_z1d):

## RESOLUTION  $z^{(1*d)}$ 

Solutions_z1d := solve(Systeme_z1d, Variables_z1d):

assign(Solutions_z1d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
```


51255
51255
51255
51255
51255
180716
331168
17799
17799
62113
95400
1932

(3)

```
> ## SYSTEM FOR DIVISIBILITY BY  $z^{(2*d)}$ 

Systemez2d := {}:

for k from 0 to (m-2)/3 do

  Coeffs_z2d[2,m-2-3*k,k] := {coeffs(expand(mtaylor(Eq[2,m-2-3*k,
k], z, 2*d)), [x,y,z])}:

  Systemez2d := Systemez2d union Coeffs_z2d[2,m-2-3*k,k]:

od:

Systeme_z2d := Systemez2d:

## VARIABLES FOR DIVISIBILITY BY  $z^{(2*d)}$ 

Variables_z2d := indets(Systeme_z2d):

## RESOLUTION  $z^{(2*d)}$ 

Solutions_z2d := solve(Systeme_z2d, Variables_z2d):

assign(Solutions_z2d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
```

```

for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;

```

51255

51255

51255

51255

473511

382072

652987

17799

23658

29916

34802

1794

(4)

```

> ## SYSTEM FOR DIVISIBILITY BY  $z^{(3*d)}$ 

Systemez3d := {}:

for k from 0 to (m-3)/3 do

Coeffs_z3d[3,m-3-3*k,k] := {coeffs(expand(mtaylor(Eq[3,m-3-3*k,
k], z, 3*d)), [x,y,z])}:

Systemez3d := Systemez3d union Coeffs_z3d[3,m-3-3*k,k]:

od:

Systeme_z3d := Systemez3d:

## VARIABLES FOR DIVISIBILITY BY  $z^{(3*d)}$ 

Variables_z3d := indets(Systeme_z3d):

## RESOLUTION  $z^{(3*d)}$ 

Solutions_z3d := solve(Systeme_z3d, Variables_z3d):

assign(Solutions_z3d):

##

```

```

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;

```

51255

51255

51255

557642

588550

571788

605476

2268

3636

3007

1248

1863

(5)

```

> ## SYSTEM FOR DIVISIBILITY BY  $z^{(4*d)}$ 

```

```

Systemez4d := {}:

```

```

for k from 0 to (m-4)/3 do

```

```

  Coeffs_z4d[4,m-4-3*k,k] := {coeffs(expand(mtaylor(Eq[4,m-4-3*k,
k], z, 4*d)), [x,y,z])}:

```

```

Systemez4d := Systemez4d union Coeffs_z4d[4,m-4-3*k,k]:

```

```

od:

```

```

Systeme_z4d := Systemez4d:

```

```

## VARIABLES FOR DIVISIBILITY BY  $z^{(4*d)}$ 

```

```

Variables_z4d := indets(Systeme_z4d):

```

```

## RESOLUTION  $z^{(4*d)}$ 

```

```

Solutions_z4d := solve(Systeme_z4d, Variables_z4d):

```

```
assign(Solutions_z4d):
```

```
##
```

```
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;  
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;  
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;  
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
```

51255

51255

188763

244239

274517

274452

249459

4113

6907

7149

5155

2116

(6)

```
> ## SYSTEM FOR DIVISIBILITY BY  $z^{(5*d)}$ 
```

```
Systemez5d := {}:
```

```
for k from 0 to (m-5)/3 do
```

```
  Coeffs_z5d[5,m-5-3*k,k] := {coeffs(expand(mtaylor(Eq[5,m-5-3*k,  
  k], z, 5*d)), [x,y,z])}:  
end do;
```

```
Systemez5d := Systemez5d union Coeffs_z5d[5,m-5-3*k,k]:
```

```
od:
```

```
Systeme_z5d := Systemez5d:
```

```
## VARIABLES FOR DIVISIBILITY BY  $z^{(5*d)}$ 
```

```
Variables_z5d := indets(Systeme_z5d):
```

```
## RESOLUTION  $z^{(5*d)}$ 
```

```
Solutions_z5d := solve(Systeme_z5d, Variables_z5d):
```

```
assign(Solutions_z5d):
```

```
##
```

```
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
```

```
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
```

```
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
```

```
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
```

51255

4114

8201

9375

9560

8824

5916

1573

2531

2497

1612

907

(7)

```
> ## SYSTEM FOR DIVISIBILITY BY  $z^{(6*d)}$ 
```

```
Systemez6d := {}:
```

```
for k from 0 to (m-6)/3 do
```

```
  Coeffs_z6d[6,m-6-3*k,k] := {coeffs(expand(mtaylor(Eq[6,m-6-3*k,  
k], z, 6*d)), [x,y,z])}:  
  Systemez6d := Systemez6d union Coeffs_z6d[6,m-6-3*k,k]:  
  od:
```

```
Systeme_z6d := Systemez6d:
```

```
## VARIABLES FOR DIVISIBILITY BY  $z^{(6*d)}$ 
```

```
Variables_z6d := indets(Systeme_z6d):
```

```
## RESOLUTION  $z^{(6*d)}$ 
```

```
Solutions_z6d := solve(Systeme_z6d, Variables_z6d):
```

```
assign(Solutions_z6d):
```

```
##
```

```
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;  
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;  
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;  
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
```

3

5

7

7

6

1

1

5

7

6

1

6

(8)

```
> ## HOLOMORPHICITY AT INFINITY
```

```
for l from 0 to m - 0*3 do
```

```
HA||l := add(add(add(A||l[i,j,k]*X^i*Z^k*Y^j*T^(dA-i-j-k), i=0..  
dA-j-k), k=0..dA-j), j=0..min(d,dA));
```

```
od:
```

```
#
```

```
for l from 0 to m - 1*3 do
```

```
HB||l := add(add(add(B||l[i,j,k]*X^i*Z^k*Y^j*T^(dB-i-j-k), i=0..  
dB-j-k), k=0..dB-j), j=0..min(d,dB)):
```

```
od:
```

```
#
```

```
for l from 0 to m - 2*3 do
```

```
HC||l := add(add(add(C||l[i,j,k]*X^i*Z^k*Y^j*T^(dC-i-j-k), i=0..  
dC-j-k), k=0..dC-j), j=0..min(d,dC)):
```

```
od:
```

```
#
```

```
for l from 0 to m - 3*3 do
```

```
HD||l := add(add(add(D||l[i,j,k]*X^i*Z^k*Y^j*T^(dD-i-j-k), i=0..  
dD-j-k), k=0..dD-j), j=0..min(d,dD)):
```

```
od:
```

```
##
```

```
for h from 0 to m - 3*0 do
```

```
HAEEq[h] := add(HA||l*binomial(l,h)*X^(m-3*0-l)*Y^(l-h), l=h..  
m-3*0):  
dYA[h] := degree(HAEEq[h], Y):
```

```
od:
```

```
#
```

```
for h from 0 to m - 3*1 do
```

```
HBEEq[h] := add(HB||l*binomial(l,h)*X^(m-3*1-l)*Y^(l-h), l=h..  
m-3*1):  
dYB[h] := degree(HBEEq[h], Y):
```

```
od:
```

```
#
```

```
for h from 0 to m - 3*2 do
```

```

HCEEq[h] := add(HC||I*binomial(I,h)*X^(m-3*2-I)*Y^(I-h), I=h..
m-3*2):
dYC[h] := degree(HCEEq[h], Y):

```

```

od:

```

```

#

```

```

for h from 0 to m - 3*3 do

```

```

HDEEq[h] := add(HD||I*binomial(I,h)*X^(m-3*3-I)*Y^(I-h), I=h..
m-3*3):
dYD[h] := degree(HDEEq[h], Y):

```

```

od:

```

```

##

```

```

for h from 0 to m - 3*0 do
for j1 from 0 to dYA[h] do

```

```

CoeffA[h,j1] := coeff(HAEEq[h], Y, j1):

```

```

od: od:

```

```

#

```

```

for h from 0 to m - 3*1 do
for j1 from 0 to dYB[h] do

```

```

CoeffB[h,j1] := coeff(HBEEq[h], Y, j1):

```

```

od: od:

```

```

#

```

```

for h from 0 to m - 3*2 do
for j1 from 0 to dYC[h] do

```

```

CoeffC[h,j1] := coeff(HCEEq[h], Y, j1):

```

```

od: od:

```

```

#

```

```

for h from 0 to m - 3*3 do
for j1 from 0 to dYD[h] do

```



```

CoeffD[h,j1] := coeff(HDEEq[h], Y, j1):

od: od:

##

for j1 from 0 to 1+m*d+d do

nY[j1] := (-Z^(d+1)-HUU)^(floor(j1/(d+1))) * Y^(j1-(d+1)*floor
(j1/(d+1))):

od:

##

for h from 0 to m - 3*0 do

HAEq[h] := expand(mtaylor(add(HAEEq[h]*nY[j1], j1=0..dYA[h]), T,
m-3*0-h)):
degree(HAEq[h],Y):

od:

#

for h from 0 to m - 3*1 do

HBEq[h] := expand(mtaylor(add(HBEEq[h]*nY[j1], j1=0..dYB[h]), T,
m-3*1-h)):
degree(HBEq[h],Y):

od:

#

for h from 0 to m - 3*2 do

HCEq[h] := expand(mtaylor(add(HCEEq[h]*nY[j1], j1=0..dYC[h]), T,
m-3*2-h)):
degree(HCEq[h],Y):

od:

#

for h from 0 to m - 3*3 do

HDEq[h] := expand(mtaylor(add(HDEEq[h]*nY[j1], j1=0..dYD[h]), T,

```

```

m-3*3-h)):
degree(HDEq[h],Y):

od:

```

```

> HSysteme :=      {seq(coeffs(HAEq[h],[T,X,Y,Z]),h=0..m-3*0)}
                    union {seq(coeffs(HBEq[h],[T,X,Y,Z]),h=0..m-3*1)}
                    union {seq(coeffs(HCEq[h],[T,X,Y,Z]),h=0..m-3*2)}
                    union {seq(coeffs(HDEq[h],[T,X,Y,Z]),h=0..m-3*3)}
minus {0}:

HSolutions := solve(HSysteme):

assign(HSolutions):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;

1
1
1
1
1
1
1
1
1
1
1
1
1

```

```

> Jetyx := factor(
+ (1/Rz^(m-0*2)) * x^(-twist) * add(AA||j * x1^(m-0*3-j) * (y1)^j
* Delta_yx^0, j=0..m-0*3)
+ (1/Rz^(m-1*2)) * x^(-twist) * add(BB||j * x1^(m-1*3-j) * (y1)^j
* Delta_yx^1, j=0..m-1*3)
+ (1/Rz^(m-2*2)) * x^(-twist) * add(CC||j * x1^(m-2*3-j) * (y1)^j
* Delta_yx^1, j=0..m-2*3)
+ (1/Rz^(m-3*2)) * x^(-twist) * add(DD||j * x1^(m-3*3-j) * (y1)^j
* Delta_yx^1, j=0..m-3*3)
) :

```

```

[ > Vars := indets(Jetyx) minus {x, x1, x2, y, y1, y2, z}:
    NbVars := nops(Vars);

                                NbVars:= 0
(10)

```

```

[ > Ens_Vars := {}:
    for i from 1 to nops(Vars) do
        Ens_Vars := Ens_Vars union {op(i,Vars)=0}:
    od:
    EnsVars := Ens_Vars:

```

```

[ > for i from 1 to nops(Vars) do
    JetSol[i] := factor(subs(EnsVars, subs(op(i,Vars)=1, Jetyx))):
od:

```

```

[ > ## VERIFICATION OF HOLOMORPHICITY AT INFINITY

    Liste_x := {x = 1/x,
                y = y/x,

```

```

        z = z/x,
        x1 = - x1/x^2,
        x2 = 2*x1^2/x^3 - x2/x^2,
        y1 = y1/x - y*x1/x^2,
        y2 = y2/x - 2*y1*x1/x^2 + 2*y*x1^2/x^3 - y*x2/x^2}:

##

for i from 1 to nops(Vars) do
  JetSol_x[i] := factor(subs(Liste_x, JetSol[i]]):
od:

```

```

> Vars := indets(Jetyx) minus {x, x1, x2, y, y1, y2, z}:
  NbVars := nops(Vars);

```

NbVars:= 0

(11)